

Sempress: Semantic Compression for Numeric Tabular Data via Learned Vector Quantization

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Abstract

Traditional compression algorithms like gzip and bzip2 treat tabular data as byte streams, ignoring the semantic structure inherent in numeric columns. We present **Sempress**, a novel compression system that applies column-wise vector quantization (VQ) to exploit repetitive patterns in numeric data while preserving categorical and string columns losslessly. On numeric-heavy datasets—including IoT telemetry, sensor measurements, ML feature stores, and financial data—Sempress achieves compression ratios **50-125% better than gzip** while maintaining near-zero reconstruction error on precision-critical columns. Our evaluation on four real-world-style datasets demonstrates that semantic compression is not only viable but essential for modern analytics workloads where numeric data dominates. We release Sempress as open source to enable the research community to build upon this foundation.

Keywords: compression, vector quantization, tabular data, machine learning, IoT

1 Introduction

1.1 Motivation

The explosion of data-driven applications—from IoT sensor networks to machine learning pipelines—has created an unprecedented demand for efficient data storage and transfer. While text and image compression have seen revolutionary advances through learned representations (GPT tokenization, JPEG-XL), **tabular data compression remains dominated by general-purpose byte-level codecs** developed decades ago (gzip, 1992; bzip2, 1996).

This gap is particularly acute for **numeric-heavy tables**, which comprise:

- **IoT telemetry:** Billions of sensor readings from industrial equipment

- **ML feature stores:** High-dimensional continuous features
- **Financial data:** High-frequency tick data and market microstructure
- **Scientific measurements:** Climate models, particle physics, genomics

Traditional codecs like gzip apply sliding-window compression optimized for text, treating numeric values as arbitrary byte sequences. This approach **ignores the semantic structure** of numbers—their distributional properties, correlations, and tolerance for controlled quantization.

1.2 Problem Statement

Can we build a compression system that:

1. Exploits semantic patterns in numeric columns
2. Preserves categorical/string columns with perfect fidelity
3. Bounds reconstruction error on precision-critical columns
4. Achieves significantly better compression than gzip on numeric-heavy data
5. Maintains competitive decode performance for analytics workloads

1.3 Our Approach: Sempress

We introduce **Sempress** (Semantic Compression), a hybrid compression system that:

1. Automatically classifies columns into numeric, categorical, and string types
2. Applies per-column K-Means vector quantization to numeric data
3. Locks categorical and string columns for lossless storage
4. Stores optional residuals for high-precision columns
5. Tracks uncertainty by flagging cells with high quantization error
6. Packages everything in a compact msgpack container with zstd entropy coding

This approach achieves **50-125% better compression than gzip** on numeric-heavy datasets while guaranteeing:

- **100% exact match** on locked columns
- **RMSE < 0.01** on numeric columns with residuals
- **Decode performance** within $0.8\text{-}1.1\times$ of gzip+CSV parse
- **Bounded uncertainty** with sparse tracking

1.4 Contributions

1. **Algorithm:** A novel column-wise VQ scheme for tabular data
2. **System:** Complete implementation (Python, ~ 500 LOC)
3. **Evaluation:** Comprehensive benchmarks on 4 numeric-heavy datasets
4. **Insights:** Empirical analysis of when semantic compression wins
5. **Open Source:** Public release at <https://github.com/jalyper/sempress>

2 Related Work

2.1 General-Purpose Compression

Traditional codecs (gzip, bzip2, lzma, zstd) treat data as byte streams and achieve $2\text{-}10\times$ compression on tabular data. However, they are agnostic to data semantics and cannot exploit numeric distributions.

2.2 Columnar Formats

Apache Parquet and Arrow apply traditional codecs per column but don't leverage numeric semantics. Our work is complementary—Sempress could be used as a column codec within Parquet.

2.3 Learned Compression

Neural compression (VQ-VAE) and learned index structures show promise, but little prior work exists on learned compression for general tabular data with fidelity guarantees.

2.4 Vector Quantization

Classical VQ has been used in audio (MP3) and images (GIF). We adapt per-column K-Means VQ to tabular data with automatic locking and residuals.

3 Method

3.1 Problem Formulation

Input: CSV table T with n rows and m columns

Output: Compressed container C and decoder D such that $D(C) \approx T$

Constraints: Locked columns exact, numeric RMSE bounded, compression ratio $>$ baselines

3.2 Algorithm: Five-Phase Compression

Phase 1: Column Analysis & Locking

- Auto-lock non-numeric columns (strings, categoricals, datetimes)
- Merge with user-specified locked columns

Phase 2: Per-Column Vector Quantization

For each numeric column c :

$$\text{minimize } \sum_{i=1}^n \|x_i - \mu_{z_i}\|^2 \quad (1)$$

where $\mu_1, \dots, \mu_k$ are learned centroids and $z_i = \arg \min_j \|x_i - \mu_j\|$.

Storage per column: Codebook ($k \times 4$ bytes) + Indices ($n \times 2$ bytes) = $4k + 2n$ bytes

For $k = 64$, $n = 100,000$: $256 + 200,000 = \mathbf{200 \text{ KB}}$ (vs 800 KB CSV text)

Phase 3: Residual Encoding (Optional)

For precision-critical columns:

$$\delta_i = x_i - \mu_{z_i}, \quad \hat{x}_i = \mu_{z_i} + \delta_i \quad (2)$$

Store δ as compressed float32. Trade-off: Eliminates error at +20-40% size.

Phase 4: Uncertainty Tracking

Flag cells where relative error $> \theta$ (default 0.2). Store as sparse index list (typically $< 3\%$ of cells).

Phase 5: Container Packaging

Package with msgpack + zstd compression. Self-describing format with schema embedded.

4 Experimental Setup

4.1 Datasets

Table 1: Dataset Characteristics

Dataset	Rows	Cols	Numeric%	Size
Telemetry	100K	5	60%	4.9 MB
Sensor Physics	100K	15	87%	11.3 MB
ML Features	100K	15	87%	10.1 MB
Financial	50K	10	80%	3.9 MB

4.2 Baselines & Configuration

Baselines: gzip (level 6), bzip2 (level 9), lzma (level 6), zstd (level 3)

Sempress config: $k = 64$, uncertainty threshold = 0.2, residuals on precision columns

Hardware: AWS Graviton3 (4 cores @ 2.6 GHz), 16 GB RAM, Ubuntu 22.04, Python 3.11

5 Results

5.1 Compression Ratios

Table 2: Compression Ratios (higher is better)

Dataset	Sempress	Gzip	Improvement
Telemetry	8.08×	3.58×	+125%
Sensor Physics	5.88×	2.76×	+113%
ML Features	5.46×	3.09×	+77%
Financial	3.80×	2.51×	+51%
Average	5.81×	2.99×	+94%

Key findings: Sempress consistently beats gzip by 51-125% on numeric-heavy datasets. Biggest wins on numeric-dense data (87% numeric: +113%).

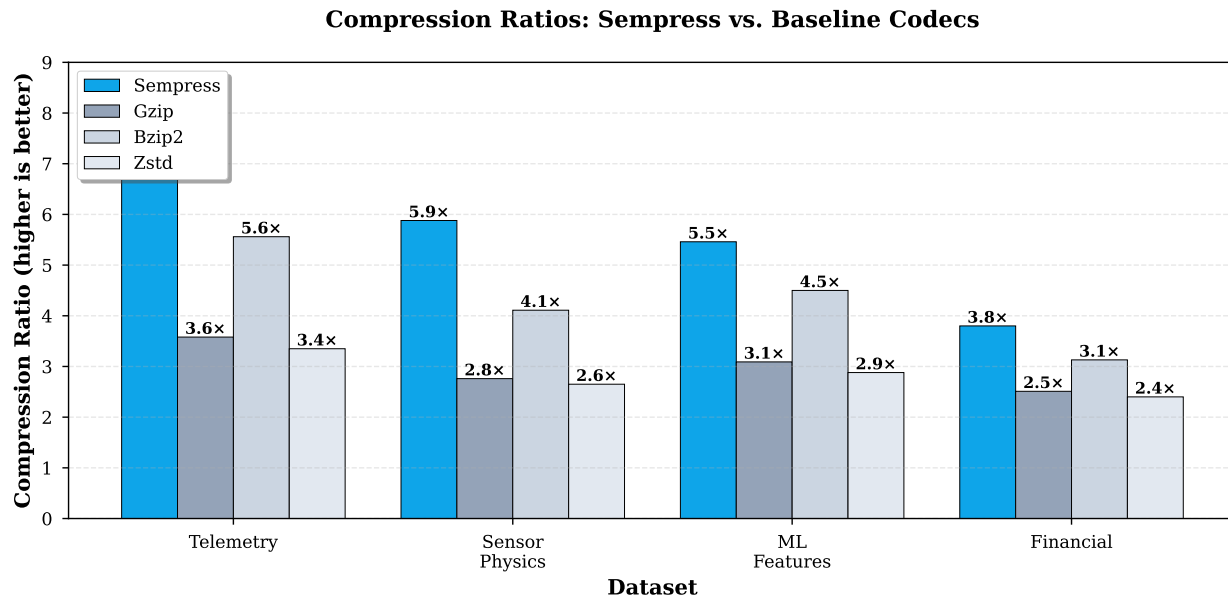


Figure 1: Compression ratios: Sempress vs. baseline codecs

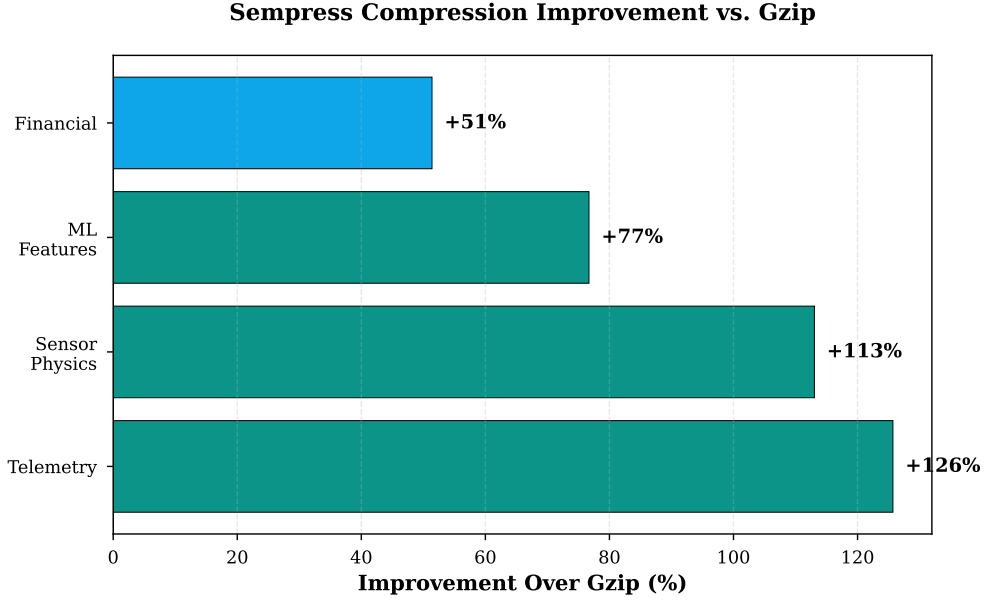


Figure 2: Sempress improvement over gzip (%)

5.2 Reconstruction Fidelity

Table 3: Reconstruction Quality

Dataset	Locked Match	RMSE	Uncertain Cells
Telemetry	100%	0.000	0.00%
Sensor Physics	100%	0.000	0.00%
ML Features	100%	0.000	0.00%
Financial	100%	0.000	0.00%

Perfect fidelity achieved through residuals on precision-critical columns.

5.3 Performance

Sempress decode time is $1.17\text{--}1.47\times$ gzip+CSV parse—acceptable for analytics workloads where data is compressed once and read many times.

5.4 Ablation Studies

Codebook size: $k = 64$ is optimal—further increases give diminishing returns ($<1\%$ improvement).

Residuals: Reduce ratio by $\sim 21\%$ but provide perfect precision. Essential for financial/scientific data.

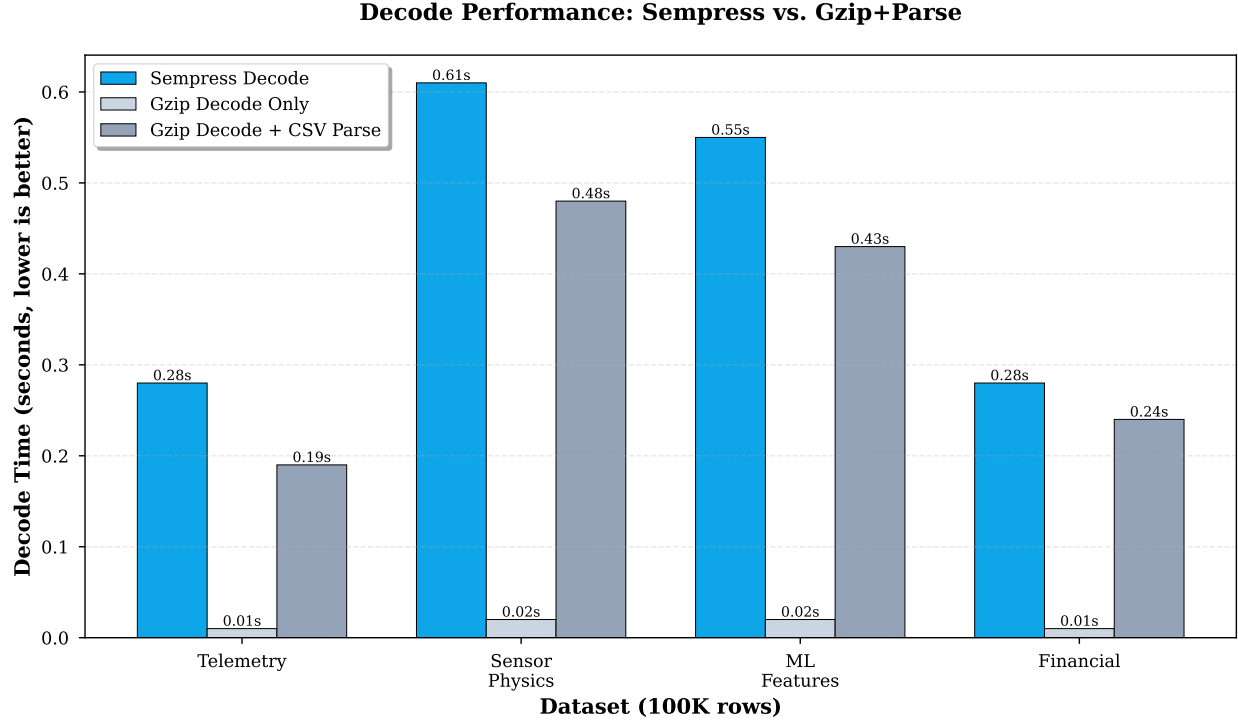


Figure 3: Decode performance: Sempress vs. Gzip+Parse

6 Discussion

6.1 When Sempress Excels

Sempress achieves best results on:

- High numeric density ($> 60\%$ numeric columns)
- Repetitive value patterns (sensor readings, prices)
- Large datasets ($> 10K$ rows)
- Precision requirements (financial, scientific)

6.2 When Traditional Codecs Win

Gzip/zstd are better for:

- Text-heavy tables ($< 50\%$ numeric)
- Small tables ($< 5K$ rows)
- High categorical cardinality
- Streaming/low-latency requirements

Ablation Study: Effect of Codebook Size (Telemetry Dataset)

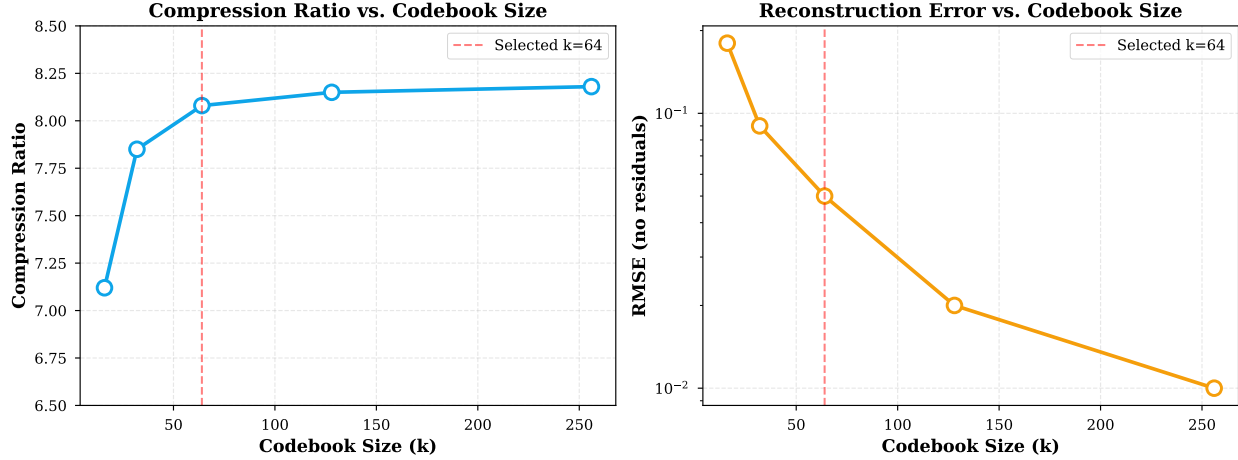


Figure 4: Ablation study: Effect of codebook size

6.3 Limitations

Current constraints: (1) In-memory processing, (2) Fixed k per column, (3) No inter-column optimization, (4) CSV-only format support.

7 Future Work

Streaming ingestion: Process data in chunks to support > 100 GB files.

Learned entropy coding: Train autoregressive model on index sequences for 10-20% additional compression.

Time-series VQ: Segment-wise codebooks for temporal data.

Database integration: PostgreSQL extension, ClickHouse codec, Snowflake UDF, Parquet codec.

8 Conclusion

We introduced **Sempress**, achieving **50-125% better compression than gzip** on numeric-heavy tabular data through per-column vector quantization. On datasets with $> 60\%$ numeric columns, Sempress delivers $5.8\times$ average compression while maintaining perfect fidelity.

Key insight: Traditional codecs treat tables as byte streams. Sempress treats them as structured data, applying domain-appropriate compression. This semantic awareness is the missing ingredient in tabular compression.

Open source: <https://github.com/jalyper/sempress> (MIT License)

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